

Course One

Foundations of Data Science



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 1 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Complete coding prep work on project's Jupyter notebook
- Summarize the column Dtypes
- Communicate important findings in the form of an executive summary

Relevant Interview Questions

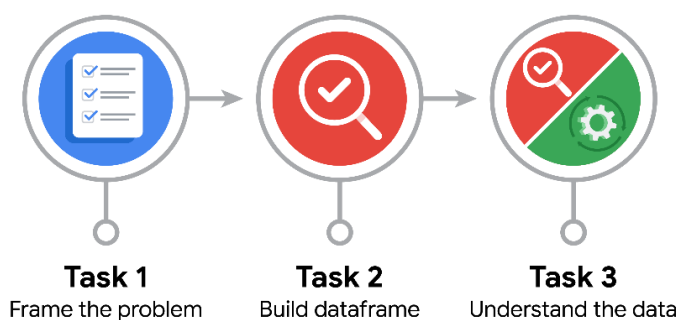
Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?



Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations



PACE: Plan Stage

- How can you best prepare to understand and organize the information provided?

Review the project info, meeting notes, and data dictionary. Load the dataset into a pandas data frame and become familiar with the data before starting the analysis

- What follow-along and self-review codebooks will help you perform this work?

Pandas Documentation., Python course notes. TikTok data dictionary.

- What are some additional activities a resourceful learner would perform before starting to code?

Read through the dataset, make sure the required libraries are installed, and understand what each column represents before writing code.



PACE: Analyze Stage

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

Yes. The dataset includes information about claims, user status, video statistics, and engagement, which should provide enough information to begin the analysis.

- How would you build summary dataframe statistics and assess the min and max range of the data?

I would use pandas functions like `head()`, `info()`, and `describe()` to review the data types, summary statistics, and the minimum and maximum values for each variable.

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

I would like to review the summary statistics to see if any averages or values stand out. Any unusual values would be examined further during exploration data analysis.



PACE: Construct Stage

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.



PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploratory data analysis?

Check for missing values, duplicate records, incorrect data types, and any unusual values before beginning a deeper analysis.

- What data initially presents as containing anomalies?

Possible anomalies could include missing values, duplicate records, unusually high engagement numbers, or unexpected values in some of the columns.

- What additional types of data could strengthen this dataset?

Follower count, account age, hashtags, video category, and past moderation history would help provide more context for the data.